



Smart City Aerial Scene Analysis for Urban Planning and Development

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80.21%



mAP: Highest accuracy using Ensemble Learning (WBF)

403,318



Annotated instances from DOTA dataset

Methodology



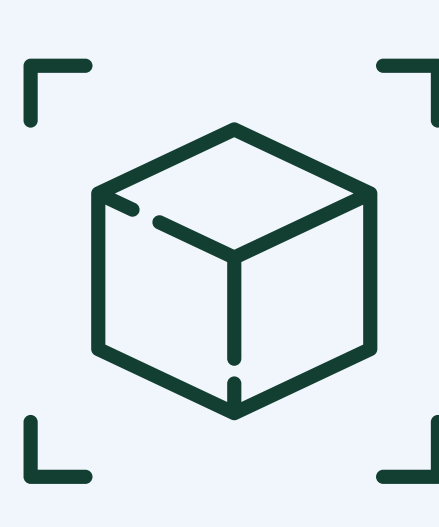
Data Collection:

The DOTA v1.5 dataset was utilized, comprising 403,318 annotated instances across complex urban categories like ships, vehicles, and infrastructure.



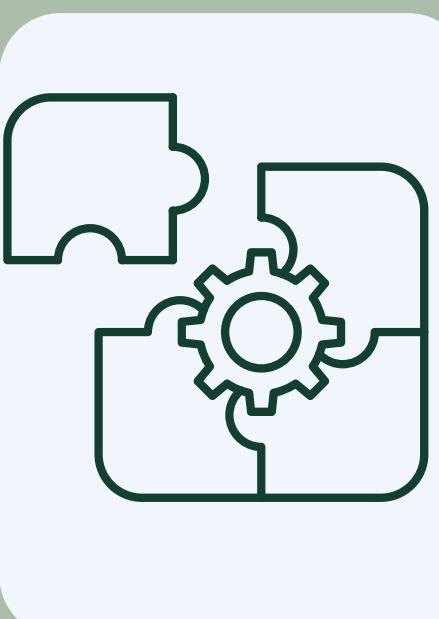
Pre-Processing:

Large images were sliced into 640x640 patches and resized to 448x448 pixels to prevent Out-Of-Memory errors. Additionally, Oriented Bounding Boxes (OBB) were converted to Horizontal Bounding Boxes (HBB) to ensure model compatibility



Object Detection:

The system processes imagery in parallel using YOLOv8 (configured for high recall) and Faster R-CNN with a ResNet-50-FPN backbone (configured for high precision)



Ensemble Fusion:

The predictions are merged using the Weighted Boxes Fusion (WBF) algorithm, assigning a higher voting weight of 3:1 to YOLOv8.

$$Coordinate_{fused} = \frac{\sum_{i=1}^n (w_i \cdot x_i)}{\sum_{i=1}^n w_i}$$

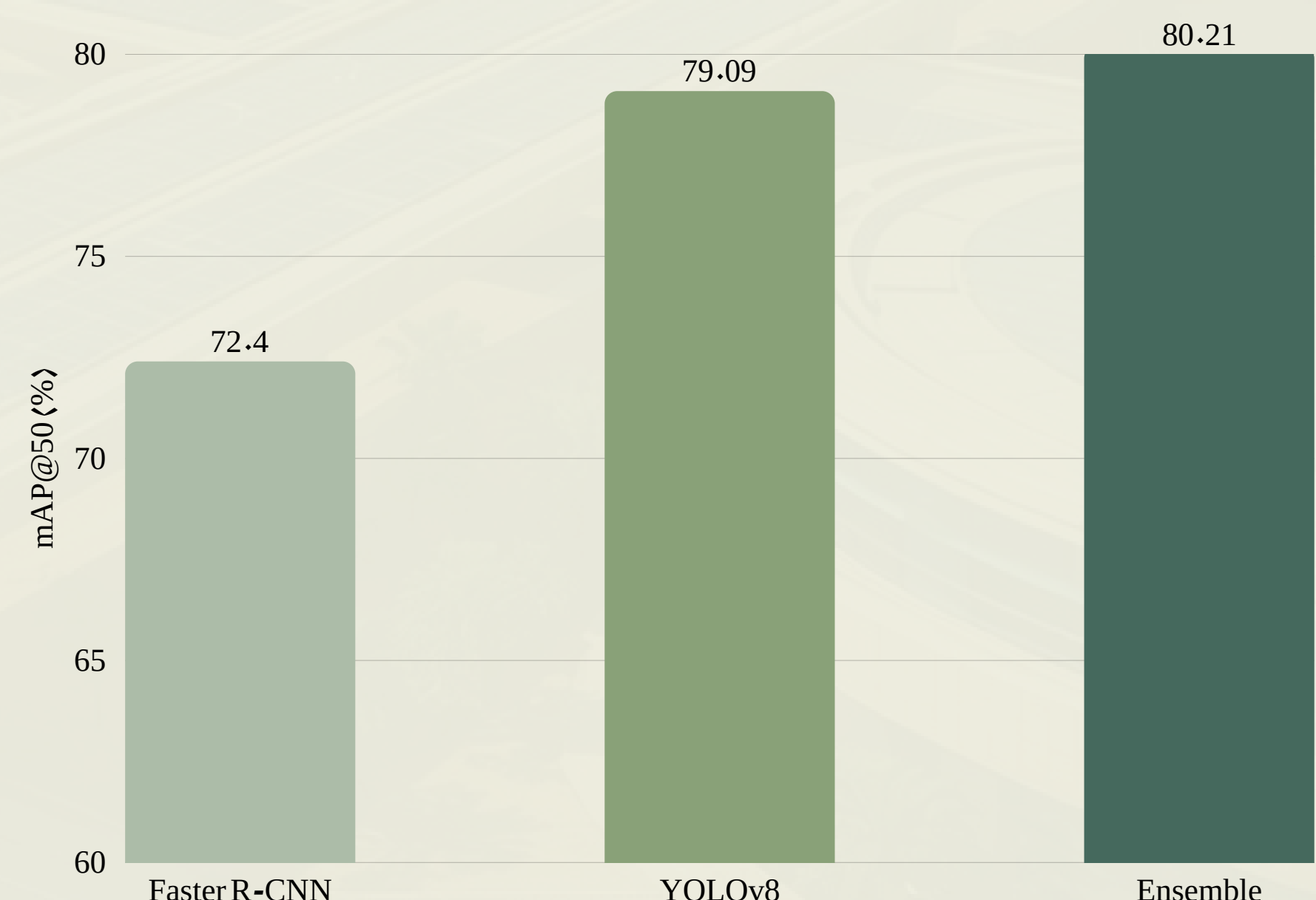
Introduction

Urban planning has traditionally relied on manual processes and limited data sources. This project addresses these inefficiencies by developing an automated Deep Learning framework for the analysis of high-resolution aerial imagery. The primary goal is to support smart city development by implementing an Ensemble Learning approach that integrates YOLOv8 and Faster R-CNN.

Results



Performance Comparison (Standardized Evaluation)



Bar chart visualizing the performance improvement. The proposed Ensemble method outperforms the single models.

Urban Planning in Action! (What's Next?)

The system offers a scalable solution for real-time aerial surveillance in complex environments, directly supporting:

Traffic Optimization: Detecting vehicles to identify congestion patterns.

Illegal Construction Monitoring: Automatically identifying unauthorized land use.

Disaster Management: Assessing critical infrastructure like bridges and safe zones for rapid response.